SQL PROJECT

TheLook eCommerce Analysis



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BACKGROUND & DATA OVERVIEW

TheLook is a fictitious eCommerce clothing site developed by The Looker team.

The Company is in the optimization mode caused by the potential crisis in 2023. The management has decided to cut off ressourcess in some categories with the lowest growth in the past 1 year.
The company would like to know the retention behaviours of the users and how to increase retention rate

The dataset contains information about customers, products, orders, logistics, web events, and digital marketing campaigns.

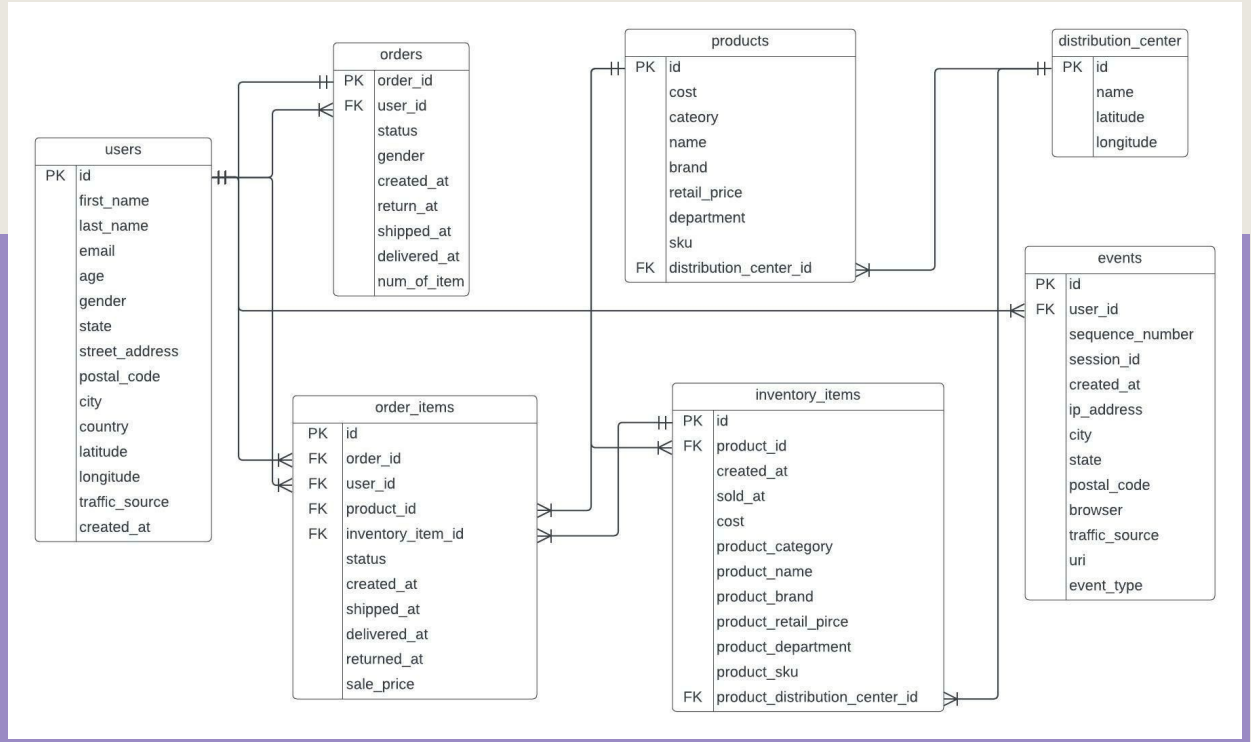
Table Of Content

1. Entity Relationship Diagram (ERD)
2. Monthly Revenue by Order Status
3. Business Questions
4. categories with the lowest revenue growth
5. categories with the lowest provit growth
6. Monthly Growth
7. Monthly Retention Cohort



ERD

ENTITY RELATIONSHIP DIAGRAM



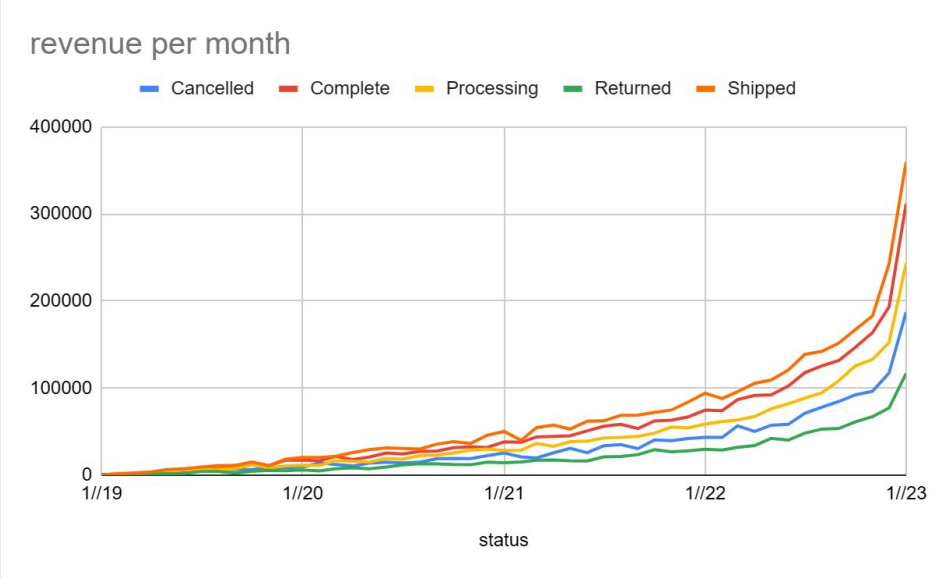
Monthly Revenue by Order Status

According to the graph, total revenues gained in all order status types are increasing month after month. This demonstrates that user activity is increasing over time.

```
SELECT
DATE_TRUNC(created_at,MONTH) month
, COUNT(created_at) order_count
, COUNT(case when status = 'Processing'
then created_at end) Processing
, COUNT(case when status = 'Cancelled'
then created_at end) Cancelled
, COUNT(case when status = 'Shipped' then
created_at end) Shipped
, COUNT(case when status = 'Complete'
then created_at end) Complete
, COUNT(case when status = 'Returned'
then created_at end) Returned

FROM
`sql-project-376612.thelook_ecommerce.ord
er_items`oi

GROUP BY 1
ORDER BY 2 DESC
```



Business Questions

1. What categories with the lowest revenue growth in the past 1 year?
2. What categories with the lowest profit growth in the past 1 year?
3. Find monthly growth of inventory in percentage breakdown by product categories, ordered by time descendingly
4. Create monthly retention cohorts, how many of them (%) coming back for the following month in 2022

What categories with the lowest revenue growth in the past 1 year?

```
WITH T1 AS (
SELECT
ii.product_category
, EXTRACT (YEAR FROM oi.created_at) year
, SUM(oi.sale_price) this_year

FROM
`sql-project-376612.thelook_ecommerce.order_items
` oi
LEFT JOIN
`sql-project-376612.thelook_ecommerce.inventory_i
tems` as ii
on inventory_item_id=ii.id
WHERE status='Complete'
group by ii.product_category, 2),

T2 AS (
SELECT *
-- , T1.YEAR
, lag(this_year) over (partition by
product_category ORDER BY year) as previous_year
FROM T1
order by 1)

SELECT *
,
((T2.this_year-T2.previous_year)/previous_year)*1
00 growth
FROM T2
WHERE year = 2022
ORDER BY growth
limit 10
```

JOB INFORMATION | RESULTS | JSON | EXECUTION DETAILS | EXECUTION GRAPH

Row	product_category	year	this_year	previous_year	growth
1	Jumpsuits & Rompers	2022	3905.64001…	2160.29999…	80.7915577…
2	Active	2022	54291.3900…	27755.6900…	95.6045408…
3	Leggings	2022	10606.1199…	5392.90998…	96.6678474…
4	Maternity	2022	32215.2499…	15622.3999…	106.211913…
5	Intimates	2022	54051.4900…	25945.4700…	108.327272…
6	Socks	2022	13983.0500…	6710.26997…	108.382822…
7	Outerwear & Coats	2022	173189.649…	80561.2999…	114.978718…
8	Pants	2022	51408.2402…	23860.9300…	115.449440…
9	Sleep & Lounge	2022	68112.4001…	31142.0700…	118.715069…
10	Dresses	2022	60105.9601…	27359.3901…	119.690423…

What categories with the lowest profit growth in the past 1 year?

```
with t1 AS (
SELECT
ii.product_id
, ii.product_category category
, EXTRACT (YEAR FROM oi.created_at) year
, count(distinct oi.order_id) total_order
, avg(ii.cost) cost
, sum (sale_price) revenue
FROM `sql-project-376612.thelook_ecommerce.order_items oi
LEFT JOIN `sql-project-376612.thelook_ecommerce.inventory_items` as ii
on inventory_item_id=ii.id
WHERE status='Complete'
GROUP BY ii.product_category, ii.product_id, oi.created_at
),
t2 AS (
SELECT category
, year
, (sum(t1.revenue)) - (sum(t1.total_order*t1.cost)) thisyear_profit
FROM t1
GROUP BY 1,2
),
t3 as (
SELECT *
, lag(thisyear_profit) over (partition by category ORDER
BY year) as previous_profit
FROM t2
GROUP BY 1,2,3)

SELECT *
,((thisyear_profit-previous_profit)/previous_profit)*100
growth
FROM T3
WHERE year = 2022
ORDER BY growth
LIMIT 10
```

Row	category	year	thisyear_profit	previous_profit	growth
1	Jumpsuits & Rompers	2022	1802.26509...	1006.24884...	79.1072952...
2	Leggings	2022	4221.60657...	2154.83639...	95.9130904...
3	Active	2022	31664.7270...	16073.3203...	97.0017787...
4	Maternity	2022	18025.7682...	8702.41303...	107.135287...
5	Socks	2022	5529.80387...	2653.19413...	108.420628...
6	Intimates	2022	25343.6781...	12138.5897...	108.786017...
7	Pants	2022	27753.7556...	12885.7278...	115.383686...
8	Outerwear & Coats	2022	96316.9253...	44449.2895...	116.689459...
9	Dresses	2022	32933.9813...	15087.9395...	118.280178...
10	Sleep & Lounge	2022	35007.6880...	15998.2584...	118.821868...

insight

- Jumpsuits & Rompies categories has the lowest sales growth this year, with a 71% increase over previous year.
- Jumpsuits & Rompies is one of the categories with the lowest profit increase, with revenue up only 75% from the previous year.

recommendation

- Product Diversification: Once the market research is complete, focus on product diversification within the Jumpsuits & Rompers category. Introduce new designs, styles, and variations that align with current fashion trends and customer preferences.
- Collaborate with influencers or designers to create exclusive collections that generate interest and drive sales. This will help attract a broader customer base and revitalize the category.
- Assess the pricing strategy for Jumpsuits & Rompers and determine if adjustments are necessary. Analyze market demand and competition to find the optimal balance between profitability and sales volume.



monthly growth of inventory

```
WITH product_count as (
  SELECT
  product_id
, product_category
, DATE_TRUNC(created_at,MONTH) month
, COUNT(DISTINCT id) count_product
FROM `sql-project-376612.thelook_ecommerce.inventory_items`
 GROUP BY 1,2,3
),

product_sold as (
SELECT
product_id
, product_category
, DATE_TRUNC(sold_at,MONTH) month_sold
, COUNT(DISTINCT id) count_sold
FROM `sql-project-376612.thelook_ecommerce.inventory_items`
WHERE sold_at IS NOT NULL
 GROUP BY 1,2,3
),

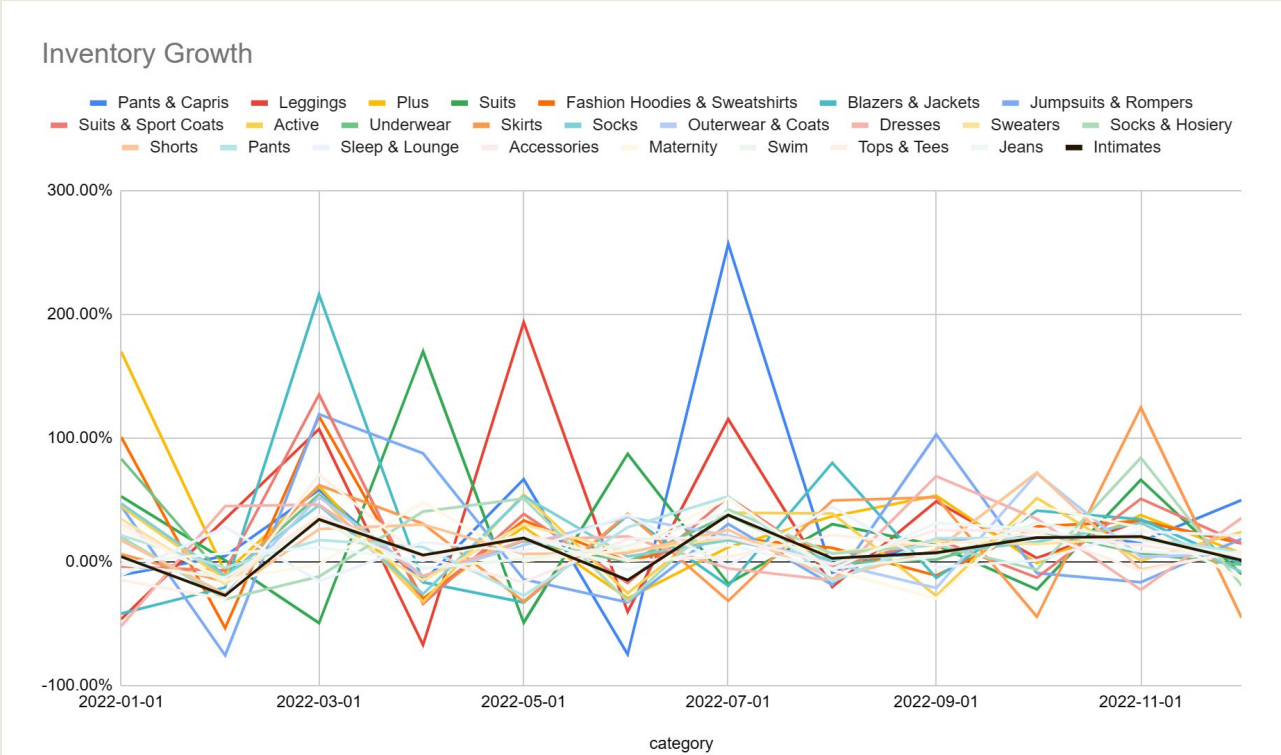
actual_product as (
SELECT
product_count.product_category as category
, EXTRACT (YEAR FROM month) year
, month
, sum (count_product-count_sold) as product
FROM product_count
JOIN product_sold
USING (product_id)
GROUP BY 1,2,3
order by 2
),

lag_product as (
SELECT *
, LAG(actual_product.product) OVER (PARTITION BY category
ORDER BY actual_product.month) last_month
FROM actual_product
group by 1, 2, 3,4 ),

growth as (
SELECT *
, (product-last_month/last_month)*100 growth
FROM lag_product
group by 1, 2, 3, 4, 5
```

Inventory Growth

AVERAGE of growth	
category	Grand Total
Pants & Capris	31.76%
Leggings	31.47%
Plus	28.16%
Suits	22.83%
Fashion Hoodies & Sweatshirts	22.20%
Blazers & Jackets	22.02%
Jumpsuits & Rompers	19.87%
Suits & Sport Coats	19.38%
Active	17.56%
Underwear	16.52%
Skirts	16.31%
Socks	14.95%
Outerwear & Coats	13.65%
Dresses	13.61%
Sweaters	13.37%
Socks & Hosiery	13.36%
Shorts	13.19%
Pants	12.58%
Sleep & Lounge	12.42%
Accessories	12.06%
Maternity	12.00%
Swim	10.22%
Tops & Tees	9.69%
Jeans	9.53%
Intimates	9.31%
Clothing Sets	-164.66%





insight

- Pants & Capris and leggings is the categories that has the biggest average growth per month in 2022
- There are categories with negative average monthly growth.

recommendation

- Allocate additional resources and attention to Pants & Capris categories to further boost sales and profitability. Increase the variety of designs, styles, and sizes available to cater to a wider customer base.
- Identify the categories with negative monthly growth and conduct a thorough analysis to understand the reasons behind the decline. Evaluate factors such as changing customer preferences, market saturation, or increased competition.

Cohort SQL Querry (1)

```sql
WITH cohort_items AS (
  SELECT
    (user_id) user_id
    , MIN(DATE(DATE_TRUNC(created_at, MONTH))) AS cohort_month
  FROM `sql-project-376612.thelook_ecommerce.orders` o
  WHERE status = 'Complete'
  GROUP BY
  user_id
 ),

user_activities AS (
  SELECT
    (user_id) user_id
    , DATE_DIFF(DATE_TRUNC(DATE(o.created_at), MONTH)
      , cohort_month, MONTH) AS month_number
    FROM `sql-project-376612.thelook_ecommerce.orders` o
    LEFT JOIN cohort_items
    USING (user_id)
    WHERE EXTRACT(YEAR FROM DATE(cohort_month)) = 2022
    AND o.status = 'Complete'
    GROUP BY user_id, month_number
),

cohort_size AS (
  SELECT
    cohort_month
    , COUNT(user_id) AS num_users
  FROM cohort_items
  GROUP BY cohort_month
),

retention_table AS (
  SELECT
    c.cohort_month
    , a.month_number
    , COUNT(1) AS num_users
  FROM user_activities AS a
  LEFT JOIN cohort_items AS c
    USING (user_id)
  GROUP BY
    c.cohort_month
    ,  a.month_number
)
```

Cohort SQL Querry (2)

```sql
SELECT
  rt.cohort_month AS cohort_month
  , cs.num_users AS cohort_size
  , rt.month_number AS month_number
  , rt.num_users AS total_users
  , ROUND( (rt.num_users/cs.num_users)*100,2) AS percentage
FROM retention_table AS rt
LEFT JOIN cohort_size AS cs
ON rt.cohort_month = cs.cohort_month
WHERE rt.cohort_month IS NOT NULL
ORDER BY

   cohort_month
   , month_number
   , cohort_size
```

QUERRY TABLE RESULT

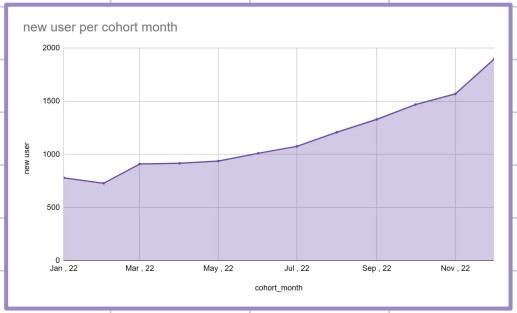
JOB INFORMATION | RESULTS | JSON | EXECUTION DETAILS | EXECUT

Row	cohort_month	cohort_size	month_number	total_users	percentage
1	2022-01-01	779	0	779	100.0
2	2022-01-01	779	1	5	0.64
3	2022-01-01	779	2	13	1.67
4	2022-01-01	779	3	10	1.28
5	2022-01-01	779	4	8	1.03
6	2022-01-01	779	5	10	1.28
7	2022-01-01	779	6	10	1.28
8	2022-01-01	779	7	8	1.03
9	2022-01-01	779	8	10	1.28
10	2022-01-01	779	9	10	1.28
11	2022-01-01	779	10	8	1.03

Monthly Retention Cohort %

AVERAGE of percentage	month_number											
cohort_month	0	1	2	3	4	5	6	7	8	9	10	11
2022-01-01	100	0.64	1.67	1.28	1.03	1.28	1.28	1.03	1.28	1.28	1.03	1.03
2022-02-01	100	1.37	1.65	0.96	0.69	1.65	1.10	1.24	0.96	1.65	0.96	0.55
2022-03-01	100	1.54	1.87	0.99	1.10	1.54	1.32	1.87	2.20	1.43	0.55	
2022-04-01	100	0.87	1.42	2.18	1.64	1.64	0.98	1.42	1.75	1.64		
2022-05-01	100	1.07	2.13	1.39	1.60	1.71	1.92	1.39	1.49			
2022-06-01	100	1.49	1.78	0.89	1.19	1.78	1.88	1.78				
2022-07-01	100	1.58	1.58	2.23	2.14	1.67	1.30					
2022-08-01	100	2.24	1.90	2.57	2.07	1.74						
2022-09-01	100	2.03	2.26	1.81	1.88							
2022-10-01	100	2.79	2.52	1.98								
2022-11-01	100	3.13	3.19									
2022-12-01	100	4.06										
Grand Total	**100**	**1.90**	**2.00**	**1.63**	**1.48**	**1.63**	**1.40**	**1.46**	**1.54**	**1.50**	**0.85**	**0.79**

new user per cohort month



INSIGHT

- The Total new user is increasing within month in 2022
- Although the total number of new users who place their first order increases each month, the overall retention percentage is quite low (3%), indicating that most of them do not make any purchases the following month.

RECOMMENDATION

- Focus on User Retention: Despite the increasing number of new users each month, the low retention percentage suggests that efforts should be directed towards retaining these users. Implement strategies such as personalized email campaigns, targeted promotions, and loyalty programs to encourage repeat purchases and build long-term customer relationships.





THANKS

Do you have any questions?

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